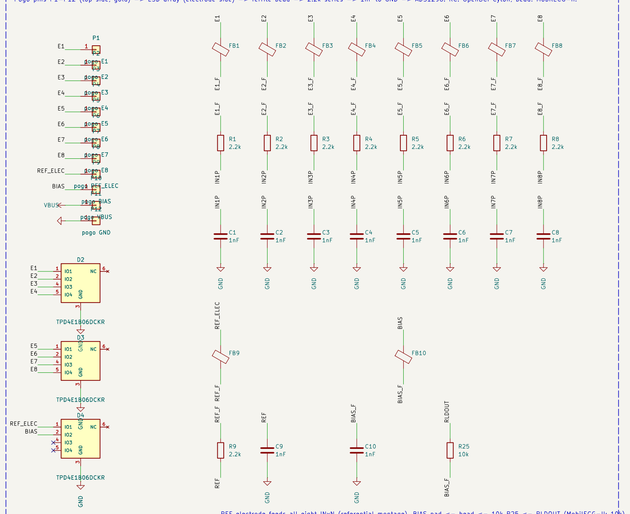
1. Electrode pogo pins + input protection

Pogo pins P1–P12 (top side, gold) -> ESD array (electrode side) -> ferrite bead -> 2.2k series -> 1nF to GND -> ADS1298. RC: OpenBCI Cyton; bead: MobiliECG-II.

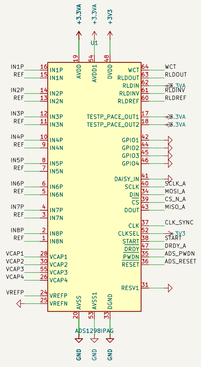


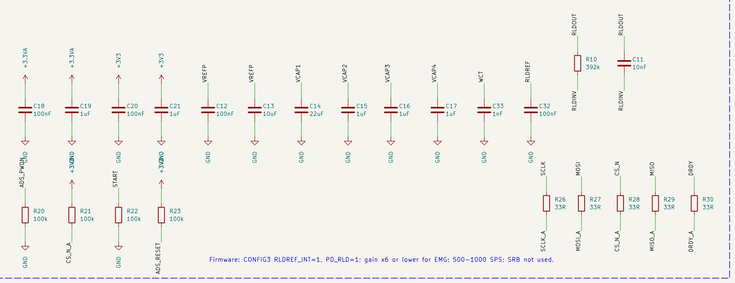
REF electrode feeds all eight INxN (referential montage). BIAS pad <- bead <- 10k R25 <- RLDOUT (MobiliECG-II: 10k).

FB1–FB10: Murata BLM15HD182SN1D (0402, GHz-band bead).

2. ADS1298 analog front end + 3. bias drive (RLD)

Unipolar: AVDD 3.3V from LDO1 only, DVDD 3.3V from LDO2. Caps per TI SBAS459K Figure 106, placed at the package.

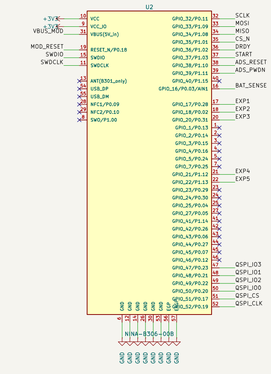


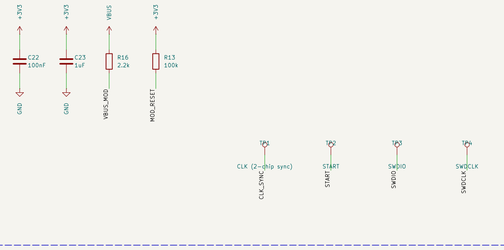


Firmware: CONFIG3 RLDREF_INT=1, PD_RLD=1; gain x6 or lower for EMG; 500–1000 SPS; SRB not used.

4. Radio: u-blox NINA-B306-00B (nRF52840, internal antenna, 32.768 kHz crystal inside)

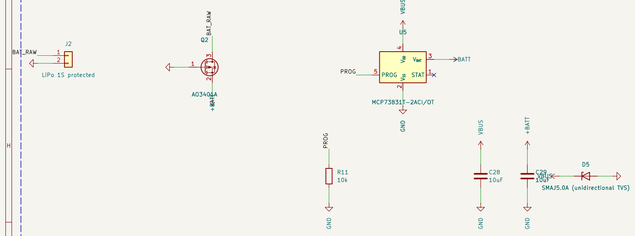
SPI on non radio-sensitive pins (P0.11/P1.09/P1.08). CS/DRDY/START/RESET/PWDN toggle < 10 kHz (u-blox note 2).

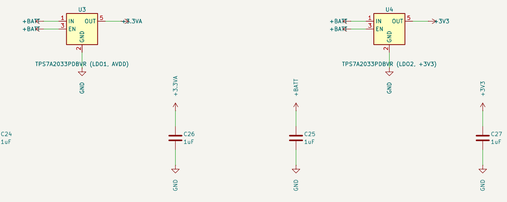




5. Power: battery, reverse-polarity FET, charger, two LDOs

Charge only off-body: firmware stops recording when VBUS is present. MCP73831 100 mA (R_PROG 10k).

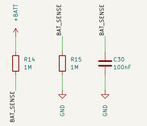






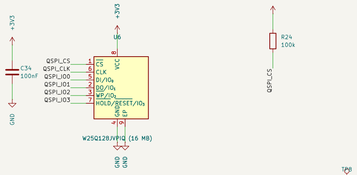
6. Battery monitor (no LEDs: hidden wearable, status shown in the phone app)

Fixed 1M/1M divider: VBAT/2 <= 2.1 V (never above the 3.3 V pin). 2.1 uA always. 100nF feeds the SAADC sample cap.



7. 16 MB flash (record without a phone) + expansion pads (open hardware: add sensors later)

W25Q128JV on the nRF52840 QSPI pins (u-blox recommended QSPI pins). EXP1–5: spare GPIOs (any can be I2C/SPI/UART/PWM; EXP1–3 also analog).



Analog front end: TI SBAS459K + ADS1298ECG-FE (SBAU171D) + OpenBCI Cyton (CC BY-SA 4.0)
Rev 0.3: + BIAS 10k, SPI 33R series, electrode ferrites (MobiliECG-II compare)

ss-pod	
Sheet: /	
File: pod.kicad_sch	
Title: ss-pod v1: ADS1298 + nRF52840 (NINA-B306) EMG pod	
Size: A1	Date:
KiCad E.D.A. 10.0.6	**Rev: 0.3**
	Id: 1/1